

42.510 Statistical Learning for Data Science

Bikramjit Das

Week 12, Lecture 2

Outline

Previous class

- Multiple testing: challenges
- Family-Wise Error Rate (FWER)
 - ▶ Bonferroni correction and Holm's method

Today's class

- False Discovery Rate (FDR)
 - ▶ Benjamini-Hochberg procedure
- Re-sampling: p -values and FDR
- R implementation

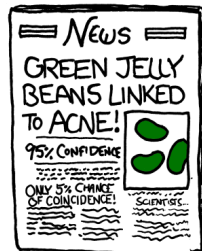
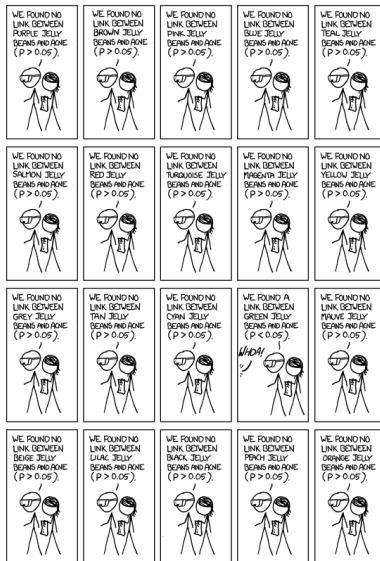
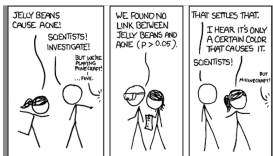
Multiple testing

- Suppose we want to test m null hypotheses (simultaneously):

$$\mathbf{H}_{01}, \mathbf{H}_{02}, \dots, \mathbf{H}_{0m}.$$

- Can we reject all hypotheses for which p -value falls below say 0.05?
- What is our Type I error here?

Multiple testing



Family-Wise Error Rate (FWER)

- The **Family-Wise Error Rate** (FWER) is the probability of making at least one Type I error when conducting m hypothesis tests.
- Then $\text{FWER} = \mathbb{P}(V \geq 1)$ in the following table of counts from the m hypotheses tests.

	H_0 is True	H_0 is False	Total
Reject H_0	V	S	R
Do not Reject H_0	U	W	$m - R$
Total	m_0	$m - m_0$	m

- If the tests are independent and all H_{0j} are true then

$$\text{FWER} = 1 - \mathbb{P}(\text{do not falsely reject any null hypotheses})$$

$$= 1 - \mathbb{P}\left(\bigcap_{j=1}^m \{\text{do not false reject hypothesis } H_{0j}\}\right)$$

$$= 1 - \prod_{i=1}^m (1 - \alpha) = 1 - (1 - \alpha)^m.$$

Bonferroni correction for controlling FWER

- Let A_j denote the event that we falsely reject the j^{th} null hypothesis. Then

FWER = $\mathbb{P}$ (falsely rejecting at least one null hypotheses)

$$= \mathbb{P}\left(\bigcup_{j=1}^m A_j\right) \leq \sum_{j=1}^m \mathbb{P}(A_j).$$

- If we only reject a hypothesis when its p -value is less than α/m then

$$\text{FWER} \leq \sum_{j=1}^m \mathbb{P}(A_j) \leq \sum_{j=1}^m \frac{\alpha}{m} = m \times \frac{\alpha}{m} = \alpha.$$

since, under null hypotheses $\mathbb{P}(A_j) \leq \alpha/m$.

- We call this the **Bonferroni correction**:
 - ▶ For controlling FWER at level α , we reject any null hypothesis with p -value below α/m .

Holm's method for controlling FWER

- Compute p -values $p_1, \dots, p_m$ for the m null hypotheses $\mathbf{H}_{01}, \mathbf{H}_{02}, \dots, \mathbf{H}_{0m}$.
- Order the m p -values so that $p_{(1)} \leq \dots \leq p_{(m)}$.
- Define

$$L = \min \left\{ j : p_{(j)} \geq \frac{\alpha}{m+1-j} \right\}.$$

- Reject all null hypotheses $\mathbf{H}_{0j}$ for which $p_j < p_{(L)}$.
 - ▶ This is called **Holm's method** for controlling FWER at level α .

False Discovery Rate

- Recall the table:

	H_0 is True	H_0 is False	Total
Reject H_0	V	S	R
Do not Reject H_0	U	W	$m - R$
Total	m_0	$m - m_0$	m

- The FWER rate tries to control $\mathbb{P}(V \geq 1)$, i.e., the probability of falsely rejecting any null hypothesis.
- This is a tough ask when m is large! It will cause us to be super conservative (i.e. to very rarely reject).
- An alternative is to try to control the **False Discovery Rate**:

$$\text{FDR} = \mathbb{E}(V/R)$$

False Discovery Rate

$$\text{FDR} = \mathbb{E} \left(\frac{V}{R} \right) = \mathbb{E} \left(\frac{\text{number of false rejections}}{\text{total number of rejections}} \right).$$

- A scientist conducts a hypothesis test on each of $m = 10000$ drug candidates.
- She wants to identify a smaller set of promising candidates to investigate further.
- She wants reassurance that this smaller set is really **promising**, i.e., not too many falsely rejected $\mathbf{H}_0$'s.
- FWER controls $\mathbb{P}(\text{at least one false rejection})$.
- FDR controls the fraction of candidates in the smaller set that are really false rejections. This is what she needs!

Benjamini-Hochberg procedure

- Specify q , the level at which to control the FDR.
- Compute p -values $p_1, \dots, p_m$ for the null hypotheses $\mathbf{H}_{01}, \mathbf{H}_{02}, \dots, \mathbf{H}_{0m}$.
- Order the m p -values so that $p_{(1)} \leq \dots \leq p_{(m)}$.

- Define

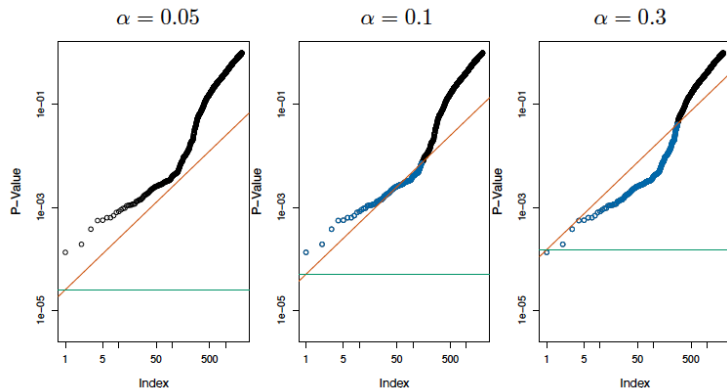
$$L = \max \left\{ j : p_{(j)} < \frac{qj}{m} \right\}.$$

- Reject all null hypotheses $\mathbf{H}_{0j}$ for which $p_j < p_{(L)}$.
- Then $\text{FDR} \leq q$.
- A control for FDR is choice different from the traditional choices of controlling FWER (0.05, 0.01, 0.001).

Example: FDR vs FWER

- We compare returns from $m = 2000$ fund managers.
- Here, p -values for $m = 2000$ null hypotheses are displayed in each panel.
- We plot controls of FWER and FDR at different levels: $\alpha = 0.05, 0.1, 0.3$.
- FWER via Bonferroni rejects $\mathbf{H}_0$ with p -values below the green line, which is 0 rejects, 0 rejects and 1 reject respectively.
- FDR via Benjamini-Hochberg rejects $\mathbf{H}_0$ with p -values below the orange line, which is 0 rejects, 146 rejects and 279 rejects respectively.
- The expected false discoveries are around $0 \times 0.05 = 0$, $146 \times 0.1 = 14.5$ and $279 \times 0.3 \approx 83.7$ respectively.

Example: FDR vs FWER



FDR vs FWER: fund manager example from previous class

- Smaller set of fund managers with $m = 5$.
- Recall the $m = 5$ p -values obtained in previous class $p_1 = 0.006$, $p_2 = 0.918$, $p_3 = 0.012$, $p_4 = 0.601$ and $p_5 = 0.756$ which were ordered:

$$p_{(1)} = 0.006, p_{(2)} = 0.012, p_{(3)} = 0.601, p_{(4)} = 0.756, p_{(5)} = 0.918.$$

- To control FDR at level $q = 0.05$ using Benjamini-Hochberg, note that

$$p_{(1)} = 0.006 < 0.05/5 = 0.01,$$

$$p_{(2)} = 0.012 < 2 \times 0.05/5 = 0.02,$$

$$p_{(3)} = 0.601 > 3 \times 0.05/5 = 0.03,$$

$$p_{(4)} = 0.756 > 4 \times 0.05/5 = 0.04,$$

$$p_{(5)} = 0.918 > 5 \times 0.05/5 = 0.05.$$

► Hence we reject $\mathbf{H}_{01}$ and $\mathbf{H}_{03}$.

- Using Bonferroni method to control FWER we only reject $\mathbf{H}_{01}$.

Resampling approach

- So far, we have assumed that we want to test some null hypothesis H_0 with some test statistic T , and that we know (or can assume) the distribution of T under H_0 .
- This allows us to compute the p -value.
- What if we do not know the theoretical distribution under the null hypothesis?

Resampling approach: two-sample t -test

- Suppose we want to test

$$\mathbf{H}_0 : \mathbb{E}(X) = \mathbb{E}(Y) \quad \text{vs.} \quad \mathbf{H}_A : \mathbb{E}(X) \neq \mathbb{E}(Y)$$

where we have n_1 samples from X and n_2 samples from Y .

- The two-sample t -statistic is

$$T = \frac{\hat{\mu}_X - \hat{\mu}_Y}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

- If n_1, n_2 are large (say, $n_1 + n_2 - 2 > 40$) then $T \sim \mathcal{N}(0, 1)$.
- If n_1, n_2 are small we do not have a hold of the theoretical distribution of T .
- Here we can try a **permutation** or **re-sampling** approach.

Resampling approach: two-sample t -test

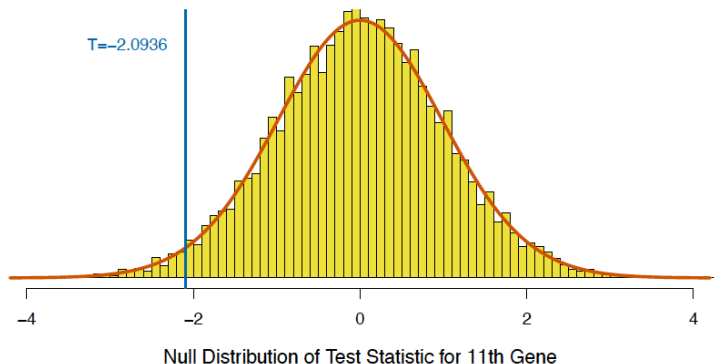
- Compute the two-sample t -statistic T on the original data:

$$x_1, \dots, x_{n_1}, \quad \text{and}, \quad y_1, \dots, y_{n_2}.$$

- For $b = 1, \dots, B$ (take B to be large, say 1000):
 - 1 Randomly shuffle the $n_1 + n_2$ observations.
 - 2 Call the first n_1 shuffled observations $x_1^*, \dots, x_{n_1}^*$ and call the remaining observations $y_1^*, \dots, y_{n_2}^*$.
 - 3 Compute a two-sample t -statistic on the shuffled data, and call it T^{*b}
- Now the p -value is given by

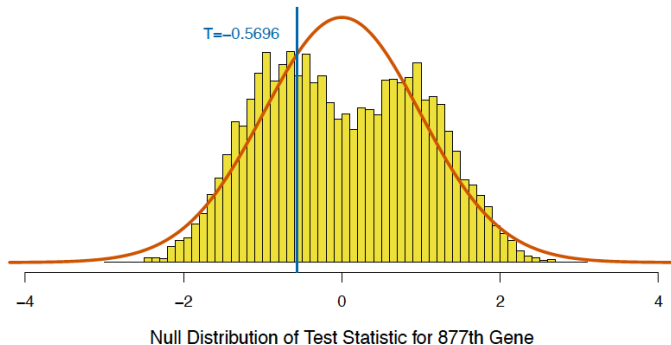
$$\frac{1}{B} \sum_{b=1}^B \mathbf{1}_{\{|T^{*b}| \geq T\}}.$$

Resampling approach: gene expression



- Theoretical p -value is 0.041. Re-sampling p -value is 0.042.

Resampling approach: gene expression



- Theoretical p -value is 0.571. Re-sampling p -value is 0.673.

Resampling approach

- Re-sampling approaches are useful if the theoretical null distribution is unavailable, or requires stringent assumptions. *So, they are actually always useful!*
- An extension of the re-sampling approach to compute a p -value can be used to control FDR.
- This example involved a two-sample t -test, but similar approaches can be developed for other test statistics.

R implementation

References

- James et al. (2023) *An Introduction to Statistical Learning: with Applications in R, Second edition*, Chapter 13.